

KARIB SHAMS

Data Scientist



✉ shams321karib@gmail.com

☎ 01797470717

📍 93 South Bashabo, Dhaka-1214, Bangladesh

🌐 github.com/karibshams

🌐 linkedin.com/in/karib-shams-007975305

🔗 karib.pythonanywhere.com

Profile

AI/ML engineer with hands-on experience in deep learning, computer vision, NLP, and RAG pipelines. Skilled in Python, TensorFlow, PyTorch, Django, and n8n automation, with a track record of leading teams to build and ship production AI systems. Backed by 17 peer-reviewed publications, including a Best Paper Award, with a sustained research focus on agricultural computer vision and graph neural networks.

Education

MSc. in CSE

East West University

01/2025 – 12/2025

CGPA: 3.91

Major in: Data Science

B.Sc. in CSE

East West University

01/2020 – 07/2024

CGPA: 3.58

HSC

National Ideal College

2017 – 2019

GPA: 4.67

SSC

Motijheel Model School And College

2016 – 2017

GPA: 5.00

Professional Experience

East West University

Graduate Teaching Assistant (GTA)

10/2024 – 12/2025

Taught Statistics, AI, and Machine Learning to 70+ students per semester; mentored 90+ students in ML projects and data analysis.

East West University

Research Assistant

10/2024 – 12/2025

Co-authored 17 peer-reviewed publications in Data Science, Computer Vision, and NLP (1 Best Paper Award). Contributed to the TFP-BD dataset, published in Data in Brief (2025).

Join Venture Ai (JVai)

Senior Executive Data Scientist | Team Leader

06/2025 – 10/2026 | Dhaka, Bangladesh

Built 13+ production AI systems (RAG chatbots, NLP tools), cutting client response time by 40%. Led a 45+ member AI team and automated n8n workflows, saving 10 hrs/week.

Professional & Academic Projects

- AI-Powered Invoice Voucher Processing System – Freelance (Fiverr): Architected a GPT-4o Vision pipeline that extracts invoice data, classifies GL accounts and profit centres, and generates payment vouchers with confidence-based review. [Link [↗](#)]
- BhromonGhuri – Full-Stack Travel & Tour Booking Platform: Built a full-scale tourism platform (Django 5, HTMX, Alpine.js) with zero-reload filtering, instant pricing, and bKash/Nagad payment verification. [Live on October]
- HealthRide – AI-driven NEMT Platform: Delivered a 24/7 GPT-4o voice receptionist and AI dispatch engine with real-time driver matching across 6 event triggers.
- Eat at Home – AI Meal Cost Estimator: Designed a GPT-4o meal-cost estimation model with store-tier blending and ZIP-code regional pricing multipliers. [Project link [↗](#)]
- OP Mental Performance AI Coach. [LINK [↗](#)]
- RAG-Based AI System: Deployed a Retrieval-Augmented Generation pipeline using FAISS embeddings and OpenAI LLMs, delivering context-aware responses across 500+ document chunks.
- Hebrew/Yiddish Handwriting Transcription System: Engineered an OCR pipeline for cursive handwriting using GPT-4o and Claude Vision APIs with CER-based accuracy evaluation. [LINK [↗](#)]
- OCR Text Extraction System: Assembled a scalable OCR engine supporting multi-format document ingestion and structured text output. [LINK [↗](#)]
- Nory0929 – Video-Based Auto Marketplace: Produced a multimodal video generation agent using Kling 2.5 and OpenAI that transforms 5+ vehicle images and an optional prompt into 25-second, 720p marketing videos.

Skills

Languages: Python, JavaScript, SQL, C, C++, Java, HTML, CSS

Frameworks: Django, FastAPI, TensorFlow, PyTorch, Scikit-learn, Streamlit

AI / ML: Deep Learning, NLP, Computer Vision, OCR, RAG Pipelines, LLMs, Generative AI, Explainable AI (SHAP), Knowledge Graphs

Automation & MLOps: n8n, LLM Chaining, Webhook Systems, OpenAI API, FAISS, Prompt Engineering

Tools: Jupyter Notebook, Google Colab, Roboflow, Kaggle, Oracle APEX, VS Code, Git

Languages

English ● ● ● ● ●
Advanced (C1)

Bengali ● ● ● ● ●
Native

Interests

Engaging in data science and analytics, traveling, and football

Awards

Received the Best Paper Award at the 5th International Conference on Applied Intelligence and Informatics (AII 2025), held in Washington, D.C., USA, for the paper titled "CodeMixEcom-Emotion: A Large-Scale Bangla-English Review Corpus and Transformer-Based Benchmark for Fine-Grained Emotion Detection."

References

Mohammad Rifat Ahmmad Rashid, Associate Professor, East West University
rifat.rashid@ewubd.edu

Musharrat Khan, Senior Lecturer, East West University
musharrat.khan@ewubd.edu

- Vehicle Detection and Traffic Flow Prediction: Trained deep learning models on the TFP-BD dataset for vehicle detection and traffic flow prediction on Bangladeshi urban roads.
- **M.Sc**  **. Thesis:** Self-Supervised Learning and Graph-Refined Object Detection Framework for Precision Agriculture
- **B.Sc**  **. Thesis:** Tuberculosis Diagnosis from Chest X-Ray Image Using Deep Learning Techniques (IEEE link )

Publications

- A. Das, K. Shams, M.R. Hasan, M.R.A. Rashid, R.U. Islam, S.H. Ripon, and A.W. Reza, "CodeMixEcom-Emotion: A Large-Scale Bangla–English Review Corpus and Transformer-Based Benchmark for Fine-Grained Emotion Detection." Proceedings of the 5th International Conference on Applied Intelligence and Informatics (AII 2025), Springer-Nature CCIS, (Accepted, 2025)
- A. Das, K. Shams, M.R. Hasan, M.R.A. Rashid, R.U. Islam, M.S.H. Khan, and R.A. Tuhin, "Towards Annotation-Efficient Kidney CT Scan Classification: Supervised and Semi-Supervised Swin Transformer Frameworks." Proceedings of the 4th IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2025), IEEE, (Accepted, 2025).
- K. Shams, A. Das, M.R. Hasan, M. Sultana, M.A. Morshed, and A.W. Reza, "Histopathology images-based deep learning prediction of prognosis and therapeutic response in small cell lung cancer." In Data Mining and Information Security: Proceedings of ICDMIS 2024, Volume 5, Springer, (2024).
- M.M. Islam, A. Das, K. Shams, M.R. Hasan, K.F. Hasan, M.R.A. Rashid, A. Chowdhury, M.S. Ali, M. Islam, M. Shahjalal, and S. Masum, "TFP-BD: An image dataset for Traffic Flow and Pedestrian movement analysis on Bangladeshi urban roads." Data in Brief, Vol. 59, (2025), P.111398
- K. Shams, R. Saha, M. I. Rodela, S. R. Simran, and M. Khan, "Tuberculosis Diagnosis from Chest X-Ray Image Using Deep Learning Techniques." Proceedings of the 2025 Fifth International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT), (2025), pp. 1–6, DOI: 10.1109/ICAECT63952.2025.10958925.
- R. Ahmed, T. T. Nova, T. F. Jarin, M. M. Hossain, K. Shams, and M. R. A. Rashid, "Real-time monitoring of oyster mushroom cultivation using CCTV and attention-enhanced ShuffleNet-based explainable AI techniques." Smart Agricultural Technology, Vol. 12, (2025), P.101571. DOI: [https://doi.org/10.1016/j.atech.2025.101571 ].
- B. Hoque, R. Uddin, K. Shams, M. M. Hossain, F. A. Tasnim, M. R. A. Rashid, R. Ul Islam, and M. S. H. Khan, "Interpretable Illness-Category Classification from Drug Attributes Using XGBoost with SHAP Explanations: A Study on the Pharma-Safe Index Dataset." Proceedings of the 2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN), (2025), pp. 1–6, DOI: 10.1109/QPAIN66474.2025.11172160.